

# CEVA-TeakLite-II DSP Core



For SMT  
CEVA-TeakLite-II

Oct. 06



## CEVA-TeakLite Background

### Leverage on the success of CEVA-Teaklite and CEVA-Oak

#### § The most established DSP cores in the industry

- § Adopted by >50 Leading OEMs and Semi companies
- § Powers more than 500M devices to date

### CEVA TeakLite DSP

#### Target Market

- § 2G wireless handsets
- § Portable audio players
- § Hard disk drives
- § Digital still cameras

#### Highlights

- § Single MAC 16-bit fixed-point processor
- § Highest code density among all CPUs & DSPs
- § Soft core, Full synthesizable

## Teaklite – Most Licensed DSP Ever

Teaklite- - Adopted by >50 Leading OEMs & Semi Companies



Teaklite- Has Powered >500M Devices to Date

## **CEVA-TeakLite-II Addresses Evolving Market Requirements**

### § High performance for DSP tasks

è 200MHz at 0.13u TSMC G, worst-case

### § Lowest cost silicon

è Only 0.5 mm<sup>2</sup> for DSP engine

è Small memory footprint, all 16-bit instruction set

### § Ultra low power consumption

è 0.06 mA/MHz for a typical DSP application

|             |                   | TeakLite-II         | ARM966E <sup>1</sup> |
|-------------|-------------------|---------------------|----------------------|
| MP3 Decoder | Memory footprint  | 14.4K byte          | 28K byte             |
|             | Power consumption | <2 mW               | 4.25 mW              |
| AAC Decoder | Memory footprint  | 20K byte            | 44K byte             |
|             | Power consumption | <2 mW               | 6.25 mW              |
| Die size    |                   | 0.5 mm <sup>2</sup> | 1.0 mm <sup>2</sup>  |

<sup>1</sup> no caches, source: [www.arm.com](http://www.arm.com)



## **CEVA-TeakLite-II Addresses Evolving Market Requirements**

### **§ Efficient control functionality**

- è Enables single processor SoCs

### **§ Increased memory space**

- è Up to 2M word for code and data

- è Allows efficient high-level coding, efficient multi-tasking and manipulation of large video data files

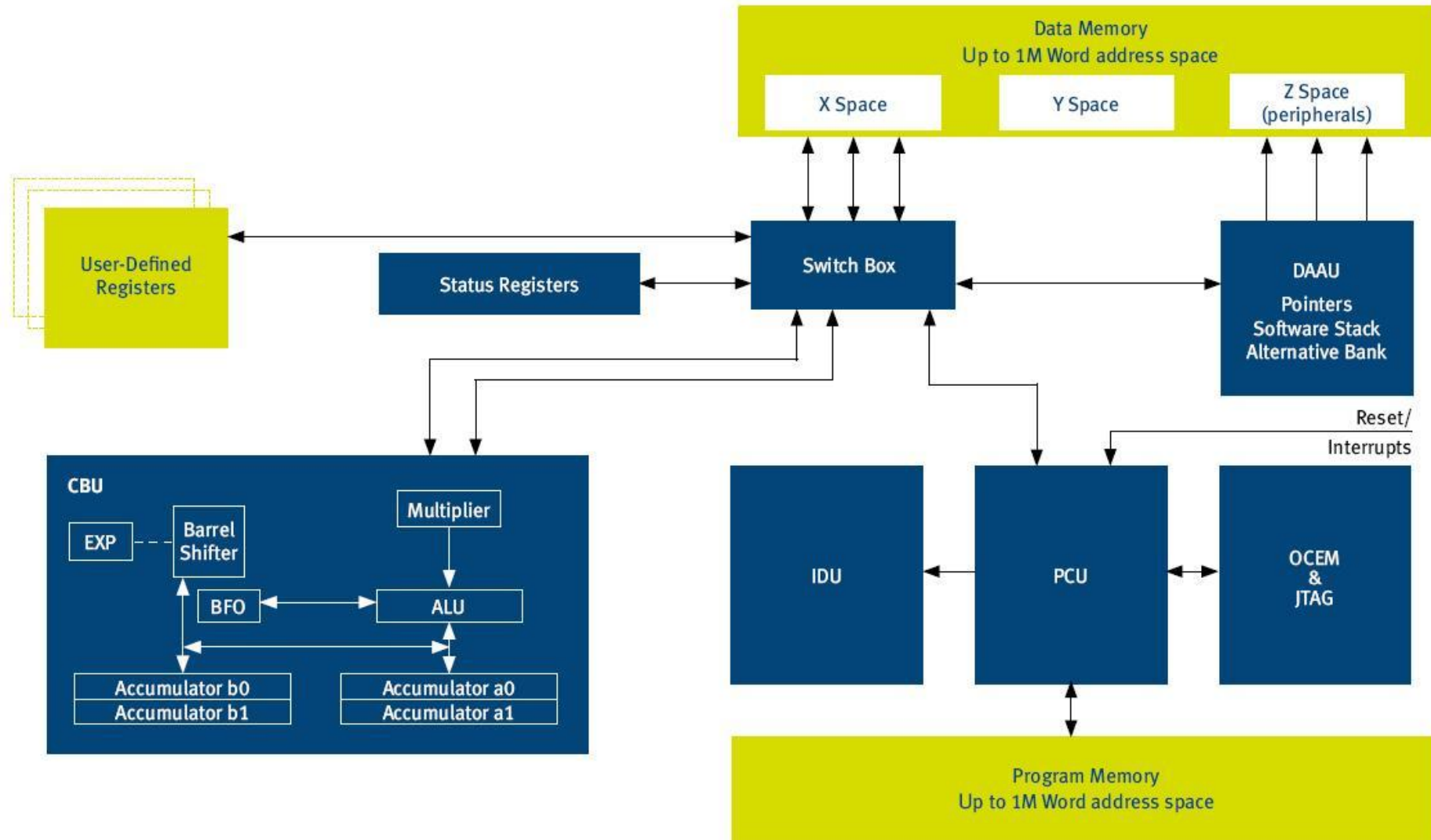
### **§ Pre-integrated with on-chip emulation and JTAG modules**

- è Eases integration and verification efforts

### **§ Seamless SW porting**

- è Backward compatible with CEVA-Oak and CEVA-Teaklite

# TeakLite-II Block Diagram



CEVA-TeakLite-II Block Diagram



## TeakLite-II – Architecture Highlights

- § **16x16-bit multiplier**
  - § Multiply-Accumulate in a single cycle
  - § Double precision multiplication
- § **Two parallel 16-bit transfers to/from data memory**
- § **36-bit Arithmetic Logic Unit (ALU)**
- § **Four independent 36-bit accumulators**
- § **Enables Two 16-bit data memory transactions in parallel**
- § **Six 16-bit general purpose address pointer registers**
- § **Enables both linear and cyclic pointer modification**





## TeakLite-II SW Development Environment



**SmartNcode™**  
Software Development Tools

- § In-house development of advanced software development environments for the last 15 years
  - § CEVA recognizes SDT importance for it's CEVA-DSPs – #1 DSP selection criteria according to “Forward Concepts”
  - § Knowledgeable support – tools developers support the tools
  - § Fast response for customer requests – CEVA's customers are 1<sup>st</sup> (and only), no need to engage with 3<sup>rd</sup> parties (SOW, contracts,...)
  - § Focus on CEVA's customers needs – we support only CEVA's DSPs
  - § Affecting CEVA-DSPs architectures to be compliant
  - § Annual new releases including new features and bug fixes
- § Mature and completely proven set of tools
- § Available on PC/Windows and Linux/UNIX/Solaris



# TeakLite-II SW Environment (Cont.)

## § Complete and Integrated SDK

## § User Friendly & advanced IDE

## § Optimized C/C++ Compiler (GNU)

## § Assembler, Linker & Utilities

## § Advanced GUI Debugger

## § Multi Core Support

## § Interface to Verilog Simulators

## § Built in Simulator

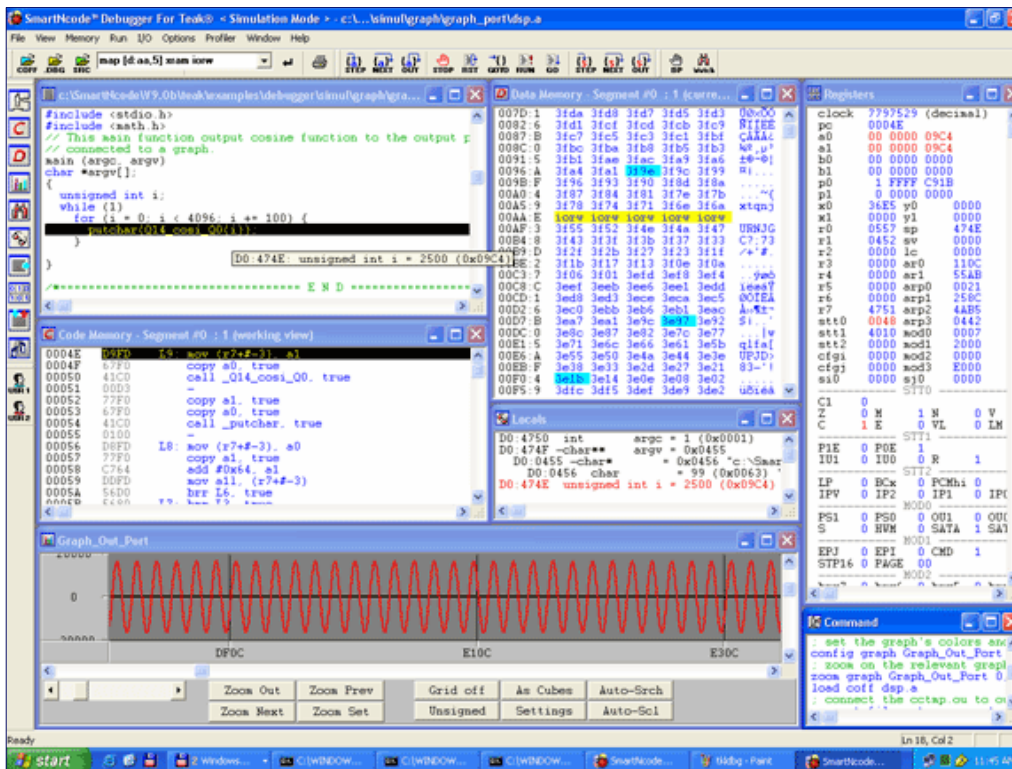
## § Instruction set

## § cycle accurate

## § Profiler

## § Performance measurement

## § MATLAB connection





## **TeakLite-II The Complete IP Solution**

- § **Fully verified synthesizable RTL code**
- § **RTL to GDSII complete and robust design flow of mainstream EDA tools**
- § **Verification and simulation environments for verifying DSP operation in a SoC**
- § **Hardware Development Tools Debug**
  - § Evaluation and Development Platform (EDP)
  - § Pre-Silicon full system and tools debug
    - § DbgVeriFier – full visibility for HW debug
    - § DSP on FPGA (same env. As EDP)
- § **Complete Software development tools chain**



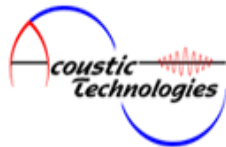
## TeakLite-II The Complete IP Solution (cont.)

- § Off-the-shelf DSP libraries and Software
- § Design services and Core Hardening
- § Comprehensive documentation
- § Technical training
- § Comprehensive technical support 24x7
- § XpertCreator solution for DSP applications integration in both, standalone DSP and DSP+RISC environments

## ➤ Established Development Community

§ Complemented by the CEVAnet – 3<sup>rd</sup> party development community

§ Application Software



§ Development tools



SYSTEMS TECHNOLOGY



§ Professional services





## CEVA-TeakLite-II Summary

- § **All new Teaklite-II DSP architecture**
  - § Based on Teaklite, the most established DSP core in the industry
- § **DSP of choice for SoCs demanding cost & power-efficient solution**
  - § 2G / 2.5G mobile handsets
  - § Portable media players
  - § Hard disk drives / servo control / optical disks
- § **Optimized performance attributes**
  - § High speed (200MHz) at a small die size (0.5mm<sup>2</sup>)
  - § Ultra low power – 0.06 mW/MHz
- § **Fully compatible with previous CEVA cores at binary level**
  - § Retains any preceding investment in SW and HW
  - § Leverages on the large SW installed based from CEVA and CEVAnet partners
- § **Enhanced functionality**
  - § Memory system, SoC integration



**CEVA-TeakLite-II**

**Thank You**

**Web: [www.ceva-dsp.com](http://www.ceva-dsp.com)**